

The Advanced Production Bible

How to build a vertical video that *demonstrates* its subject rather than captioning it — the figure catalogue, the measured colour system, the mark specs, the sound contract, the script schema, and the verification chain that refuses anything which does not comply.

THE ONE RULE THIS DOCUMENT EXISTS TO ENFORCE

No video is words on a colour field. Every video draws its subject in the middle third of the frame — a figure, a mechanism, a page being broken — and that figure is a decision written into the script, not a by-product of a template.

This is not a style guide. Every rule below is either enforced by a program that fails the build, or is a measurement with the measurement printed beside it. Where an approach was tried and rejected, the numbers that rejected it are given, so the rule can be argued with rather than merely obeyed.

- I **The exhibit doctrine** — the failure it replaces, the band, the two families
- II **The figure catalogue** — 17 kinds, what each is for, and its data contract
- III **Colour, measured** — the colourblind audit, the eight validated pairs, the ramp
- IV **Marks, scale and layout** — why a 24px bar is wrong here, and what replaces it
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- VI **Sound** — no voice, low level, and how both are made unfalsifiable
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Applies from campaign week 33. Weeks 31–32 predate the rule and are exempt by design — 31 is published and frozen, 32 is two templates that already spend their whole runtime wrecking a real page.

I · The exhibit doctrine

The failure this replaces

The templates were type on a colour field. A `SiteRoast` said “slow first view” in large letters and never showed a page. A `TechTip` listed three steps and never performed one. That reads as a caption card, and a caption card is what a viewer scrolls past: the frame carries no evidence that anything is happening, so there is nothing to stop on.

Worse, it is invisible to every check. A caption card is a perfectly healthy render — the type animates, the variance is high, the end card is there, the file is the right length with sound on it. Nothing failed. That is why the rule needed a gate of its own rather than a note in a style guide.

The band

Every template stacks the same way: hook at the top, **exhibit in the middle**, payoff at the bottom. The middle is a shared constant, not a per-template guess:

```
src/exhibits/registry.json → band: { top: 0.30, bottom: 0.74 }
```

It is one number because the verifier samples exactly those pixels. A band the layout did not actually honour would be a check passing on the wrong part of the frame — the most expensive kind of green tick. The top edge clears a three-line hook; the bottom edge clears the payoff block and the safe area the platforms overlay their own UI onto.

Two families, and why the distinction is load-bearing

CHARTS

Marks encode a measured quantity. Every length, angle and shade is $\text{value} / \text{scale}$. A chart-family figure **requires real numbers in the script** and the validator refuses prose in a numeric field.

DIAGRAMS

Marks depict a mechanism the script already states, and take their labels from the script’s own lines. Nothing is measured, so nothing is claimed.

This split exists because of a component that was deleted from this codebase. `TechTip` once drew three bars whose heights were $\text{progress} * (0.76 + \text{index} * 0.12)$ — a number derived from the frame counter and nothing else. It looked like a chart and measured nothing, which is *worse than no chart*: it invites the viewer to read a value that does not exist. The lesson is encoded as a type, not a convention.

COROLLARY — THE HONESTY RULE

If the script has no measurement, it does not get a chart. It gets a diagram, which is an equally strong picture and an honest one. Reaching for a bar chart to “look data-driven” is the failure mode, not the goal.

Where the figure comes from

1. **Authored.** `props.exhibit` in the script. Always wins. This is the intended path.
2. **Assigned.** `scripts/backfill-exhibits.mjs` writes one into the source scripts, choosing per day so the four posts six hours apart never draw the same picture. Its output is a file a human reads and edits.

3. **Derived.** `src/exhibits/derive.ts` builds one at render time from the script's own props. A floor, not a path — the validator requires an exhibit from week 33 on, so this is never reached in production.

The three cannot disagree: once the backfill has written a figure, the validator requires it and the derivation is unreachable. All three walk the same preference order in `registry.json`, and a unit test asserts the preference list and the per-kind permissions agree — the first version of this system silently stopped choosing the browser stage for `SiteRoast` because those two lists had drifted apart.

II · The figure catalogue

Seventeen kinds. The catalogue is `src/exhibits/registry.json`, and it is the only copy — the renderer, the plan validator and the pixel verifier all import it rather than keeping mirrors.

Charts — require measurements

Kind	Form	Use it when · required data
dial	Ring gauge, one ratio against its whole; the number counts inside.	One share against a limit. A two-slice pie is the wrong answer — the eye compares angles badly. value · unit · caption
bars	2–4 columns from one baseline, rounded data-end.	Magnitude, low to high. One scale, one baseline, never a second axis. series[<code>{label,value}</code>] · unit
meters	Horizontal meters, one row per item.	The categories are sentences. A column cannot carry “Headaches by mid-afternoon” at any readable size. rows[<code>{label,value}</code>] · unit
compare	Two concentric arcs and the gap between them.	Before against after. One hue in two steps — lightness survives every form of colour blindness. from <code>{label,value}</code> · to <code>{...}</code> · unit
board	Three or four tiles, exactly one emphasised.	Headline figures with no shared scale. Bars would invent a comparison the numbers do not have. tiles[3–4] · emphasis
cartogram	Tile grid map, one cell per region, one sequential hue.	Magnitude across places. A tile grid, never a projection — equal area per region, and no false geographic precision. label · unit · regions[<code>{label,value}</code>]

Diagrams — take the script's own lines

Kind	Form	Use it when · required data
pipeline	Stages with a packet travelling through, stalling at the bottleneck.	The subject is a sequence and where it waits. The packet is why it is a video and not a slide. <code>stages[3-5] · bottleneck?</code>
trace	Rows filling in under a climbing counter.	The subject is repetition. Repetition is legible only when you watch it repeat. <code>rows[3-7] · counterLabel</code>
checklist	Steps that run, hesitate, then resolve pass / warn / fail.	A printed checklist is a caption; one that resolves is a demonstration. The pause before each verdict is the device. <code>steps[{label, verdict}]</code>
nodegraph	Lit core, labelled satellites landing on an orbit.	One thing feeding many. Short labels only — ≤ 26 chars, or the layout picks a different figure. <code>core · nodes[3-6]</code>
timeline	Editor tracks, clips, a playhead crossing them.	Sequence or elapsed time. The playhead is a clock read without a number. <code>tracks[{label, clips}] · restAt?</code>
radar	Polar lattice, sweeping arm, targets appearing as it passes.	Coverage and blind spots. Targets are <i>found</i> , not listed. <code>label · targets[2-4]</code>
code	Editor panel; one marked line types, the rest appear.	The subject is literally code. Typing only the marked line puts the pace where the attention should be. <code>filename · lines[2-8] · highlight</code>
browser	The page under review, faults pinned to the part at fault, then fixed.	<code>SiteRoast</code> 's native stage. A roast about structure must show the structure.
terminal	A shell that types its command and answers it.	<code>DevJoke</code> , when the joke's setting is a machine.
chat	A message thread arriving one message at a time.	<code>DevJoke</code> , when the setting is a person. "Make the logo bigger" in a terminal is a frame arguing with its own script.
sitemock	A real page wrecked round by round.	<code>LogoLadder</code> and <code>WorksOnMyMachine</code> , which are built entirely around it.

A WORKED EXAMPLE OF EVERY KIND

`docs/exhibit-examples.json` carries one validated `exhibit` block per kind, with the template it is valid on. Paste one into a script's props and replace the data.

Five kinds — `compare` , `meters` , `cartogram` , `nodegraph` , `timeline` — are **used by none of the current 120 scripts**, because that copy carries neither the measurements nor the short labels they need. That is the honesty rule doing its job, not a gap: inventing figures to spread the catalogue evenly is the failure this system exists to prevent. They are built, rendered and tested, and the examples file is how a future script reaches them. A unit test re-validates every example, so none can rot.

CHOOSING – IN ORDER

1. Does the script state a **measurement**? → chart. Which one is decided by the job: one ratio → dial ; magnitudes → bars ; sentence-length categories → meters ; a change → compare ; unrelated headline figures → board ; places → cartogram .
2. Otherwise, what **shape** is the subject? Sequence → pipeline / timeline . Repetition → trace . Verification → checklist / radar . One-to-many → nodegraph . Code → code .
3. Does the template have a **native stage** that already is the subject? Use it.
4. Check the labels **fit**. A figure full of ellipses is worse than a different figure.

III · Colour, measured

Colour is assigned by the *job* it does, and the assignment is **computed, never eyeballed**. Four jobs, one rule each:

Job	Rule
Identity	Categorical. The validated pair below, in fixed order. Never cycled, never generated.
Magnitude	Sequential. One hue, dark → light. Never a rainbow.
Polarity	Diverging. Two poles, neutral midpoint. Not used in this series yet.
State	Status. Reserved for pass / warn / fail, and always shipped with a glyph <i>and</i> a word.

The audit that changed the palette

The eight palettes pair a warm paper ink with one saturated signal colour. Those two were chosen to look good together *as type*, where shape carries the meaning. The moment two marks in one figure are told apart by colour alone the requirement changes: they must stay distinguishable to a red–green colourblind viewer — roughly one man in twelve scrolling past. Measured with the OKLab CVD model, three of eight did not:

Palette	amber	seaglass	worst ΔE	Verdict
plum	 #FF9EC4	 #9BE3C9	4.8	FAIL — one colour to a deuteranope
moss	 #B7E36B	 #FF9F68	6.5	below the safe floor
oxblood	 #FF7A9C	 #8ED8B5	7.2	below the safe floor

Re-tinting the type was not available: weeks 31–32 are rendered and fingerprinted, and a palette edit would change what a re-render of those ids produces — which is exactly the alarm the duplicate detector exists to raise. **So charts got their own pair.** Every one clears ΔE 8 across deutan, protan and tritan, and 3:1 contrast against its own ground.

Palette	chart.primary	chart.secondary	worst ΔE	Note
aubergine	 #FFB25C	 #6FD3BE	12.0	unchanged
ink	 #7FB2FF	 #FFD166	21.5	unchanged
moss	 #B7E36B	 #FFA8E8	14.2	re-stepped to orchid
rust	 #FF8A5B	 #7FD1C4	12.9	unchanged
slate	 #F2C14E	 #8FD3F4	19.9	unchanged
plum	 #FF9EC4	 #CFE86B	15.3	re-stepped to citron
deepSea	 #5BD1C4	 #FFC46B	11.8	unchanged
oxblood	 #FF7A9C	 #8FD3F4	12.4	re-stepped to sky

ENFORCED

A plan may retint the *type* through `props.theme`. It may **not** touch `theme.chart` — the field is reinstated after the override spread in both `resolveTheme` and `lookFor`. Retinting type is a taste decision; retinting the marks that encode data would let a script hand a figure two colours measuring ΔE 4.8 apart with nothing noticing.

The sequential ramp

The obvious implementation is to fade the signal colour toward the ground. It is wrong twice, and both corrections were measured.

It is not one hue. These grounds are chromatic on purpose, so a step halfway between an orange signal and a purple ground is a muddy brown — the ramp measured **45° of hue spread** on aubergine. A ramp whose hue moves is one the reader must decode instead of read. Scaling every channel by the same factor is exactly a lightness change and leaves hue alone: measured spread is now **1°**.

Its floor was invisible. At a 0.18 floor the lowest cell measured **1.45:1** against its ground — a region carrying real data, rendered invisible. `RAMP_FLOOR = 0.5` is the smallest floor clearing 2:1 on all eight grounds; the tightest, oxblood, lands at 2.30:1.

```
rampStep(signal, t) = signal × (0.5 + t × 0.5)    // hue-preserving, visible at t=0
```

Text never wears the data colour

Marks carry the series colour. Labels, values and axis text wear ink tokens (`paper` / `paperDim`) — a light categorical hue is illegible as type, and identity is meant to come from the coloured mark *beside* the words. The single exception is a label sitting *inside* a filled mark, where the choice is made by measuring the fill's relative luminance with WCAG's own formula, not by thresholding the value that produced it.

IV · Marks, scale and layout

The scale factor — why the published mark specs are wrong here

Standard chart specs — a 24px bar cap, a 2px line, an 8px marker, a 2px gap — are written for a chart on a desktop screen where a CSS pixel is a pixel. This frame is 1080 wide and is watched on a phone roughly 390pt across, so a mark drawn at its literal spec arrives on the viewer’s eye at about a third of its intended size: a 24px bar becomes 9 points of ink, which is a hairline.

```
MARK_SCALE = 1080 / 390 = 2.77 // src/exhibits/Figure.tsx
```

Mark	Reference	This frame	Rule
Bar / column thickness	24	66	Cap it. The leftover in the band is air, not more bar.
Line weight	2	6	Round join and cap.
Marker / end-dot	8	22	Plus a surface ring where it overlaps.
Surface gap	2	6	One consistent width across a whole figure.
Hairline grid / axis	1	3	Solid, never dashed. Always recessive.
Rounded data-end	4	11	Rounded at the tip, square at the baseline.

ONE DOCUMENTED DEVIATION

bars replaces the absolute 66px cap with a proportional one — a bar takes at most **62% of its band**. The absolute cap assumes a chart sharing a dashboard with five others. Here the figure is the frame, so the proportional rule preserves the relationship the spec is actually protecting (“the leftover is air”) while letting the marks read on a phone.

The two spacers

Never draw a border around a mark to separate it. A stroke adds ink that is not data. Two mechanisms do the separating: a **surface gap** between touching marks, and a **surface ring** on dots and end-markers that cross something.

Labels — and why the usual advice is inverted here

The standard rule is *label selectively*: label the endpoint, the extreme, the one series the story is about, and let the tooltip and table view carry the rest. That rule assumes a reader who can hover.

VIDEO AMENDMENT

There is no tooltip, no legend box, no table view and no hover. The direct label beside a mark is the *only* channel left when colour fails, so **every mark is labelled**, and the figure’s panel header names what is being measured. A legend box would restate the header and cost the space the marks need.

Hard prohibitions

- **Never a second y-axis.** Two measures of different scale are two figures, not one figure with two rulers. This is the single most common way a chart lies.

- **Never generate a hue** for an extra series. Fold the tail into “other”, facet, or cut series.
- **Never colour by rank.** Colour follows the entity. A filter that changes the series count must not repaint the survivors.
- **Never clip a label** with `overflow:hidden` . If it does not fit, move it outside the mark or choose a different figure.
- **Never mark state by colour alone.** Verdicts ship with a glyph and a word.

V • Motion

Arrival, not appearance

A figure's panel rises and settles on a spring rather than fading. A fade reads as a slide transition; a physical arrival reads as an object entering the frame, and that is the difference between a deck and a video. The marks follow the panel, and a counting number rides the *same* spring as the mark it labels — a value that finishes before its bar does reads as two unrelated animations.

Motion is a variation axis

Every campaign video walks an $8 \times 7 \times 6 = 336$ -**combination** space of palette $\times$ typeface $\times$ motion signature with a stride coprime to the space, so 120 videos are distinct *by construction* rather than by luck. A signature changes entrance direction, stagger, spring damping and mass, and the backdrop's grid scale and skew — so two videos sharing a palette and a typeface still do not produce matching frames. The figure kind rotates on the same ordinal, so consecutive videos on one template never draw the same picture.

The performance budget is real

These render on a two-core runner through a software GL path. The rules are not stylistic:

- **No full-frame blur.** Glows are radial gradients. A full-frame `filter: blur()` costs more than every other layer combined.
- **Rasterise once, translate thereafter.** Animating a gradient's *position* repaints 1080×1920 twice a frame: measured $0.43 \text{ s/frame} \rightarrow 1.21 \text{ s/frame}$. Animate `transform` so each frame only composites.
- **Grain is a 256px tile, not a full-frame filter.** An `feTurbulence` over two million pixels was roughly three quarters of total render time.
- **Line work is SVG.** The rasteriser is good at it.
- **No `Math.random` in a render.** Every variation is a pure function of the video id, because renders are retried and a retry that produced a different video would invalidate the stored fingerprint.

REACT RULE THAT BITES HERE

Every figure draws a *list* of marks, so timing is computed inside a `map`. Read the frame **once** at the top of the component and pass it down to pure helpers (`arrival(theme, from, frame, fps)`). A hook called in a loop is a hook whose call order changes with the data.

VI · Sound

With no voiceover, the audio track is not decoration — it is the performance. Two requirements, and both are made unfalsifiable rather than merely stated.

No voice

Every sound in the series is **synthesised by this repository**: bed layers and one-shot cues built by `scripts/audio/`. A cue names a generated one-shot, never a file.

```
// scripts/exhibits.mjs → voiceProblems()
props.voiceover | narration | vo | speech | tts → rejected
score.cues[].sfx containing "." "/" or "\"      → rejected (that is a file, and a
                                                    file is audio this repo did not make)
```

The check is on the script, not the render, because by render time the decision has already been made. It is the only way a voice could reach the mix, so it is closed at the only place it can be closed.

Low volume — a band, not a floor

The master pass targets **−23 LUFS** at −2 dBTP. Verification checks a *band*, measured as mean RMS in dBFS:

```
QUIET_FLOOR_DB = -45    // below this: a track with effectively nothing on it
QUIET_CEILING_DB = -14   // above this: an unmastered mix at full level
observed on real mastered renders: -29.4 dB
```

The ceiling is the important half and the one a floor-only check misses entirely: it catches an unmastered mix going out at full level under a viewer's thumb at two in the morning. Measured as RMS rather than LUFS because that needs no filter build — Remotion ships `ffmpeg` with `--disable-filters` and `volumedetect` is not among the few re-enabled.

The score model

- **Bed** — layers that enter and leave on named frames. One layer per beat is the house pattern.
- **Cues** — one-shots on exact frames, with optional pitch escalation. Pitch rising across a build is this series' substitute for narration.
- **Hard silence** — cut every layer for a full beat before a payoff. Used on nearly every script. This is why the frame verifier reports stillness rather than judging it: deliberate stillness is a device here, and a check that failed it would fail the videos following the brief most closely.
- A script that transcribes no score gets the template's documented bed behaviour, so no video is ever silent by accident.

VII • The script schema

Scripts are authored in `plan-source/campaign/w<n>.json`. Ids, day numbers, eyebrows, runtimes and the week block are all *derived* by `scripts/build-campaign.mjs` — never written by hand, because that is what validators check hardest and humans are worst at.

AUTHOR HERE, NOT IN PLANS/

`plans/<week>.json` is generated. A figure written there is dropped the next time the campaign is built. The validator catches this, but the habit is worth having.

A complete item

```
{
  "template": "TechTip",
  "sourceId": "c30-d01-hero-weight",
  "caption": "Your largest paint is almost always the hero image. Three checks that
tell you in under a minute.\n\n#WebPerformance #CoreWebVitals #FrontEnd",
  "props": {
    "hook": "Your hero image is the wait",
    "steps": [
      "Open DevTools, throttle to Fast 3G",
      "Find the Largest Contentful Paint element",
      "If it is the hero, resize and preload it"
    ],
    "result": "Most sites lose two seconds to one image.",
    "variant": "vitals",
    "kicker": "SAVE THIS",

    "exhibit": {
      "kind": "checklist",
      "title": "three checks, in order",
      "steps": [
        { "label": "Throttle to Fast 3G",          "verdict": "pass" },
        { "label": "Find the LCP element",         "verdict": "pass" },
        { "label": "Hero is 1.9 MB, unpreloaded", "verdict": "fail" }
      ]
    }
  }
}
```

Rules the validator enforces on every item

Rule	Why
<code>props.exhibit</code> present (week $\geq$ 33)	The renderer would derive one anyway — but then nobody chose it. The point of the rule is that somebody chose.
Kind is known and permitted for that template	A kind no component can draw would render an empty band on a video nobody watched before it posted.
Every <code>requires</code> field present	A dial with no number is not a dial.
Chart-family <code>numeric</code> fields are finite numbers	The deleted-component rule. Prose in a numeric field means the figure should be a diagram.
List lengths inside bounds	Past them the figure stops being readable at phone size.
No two of a day's four posts draw the same figure	Four dials in one day is one video posted four times to anyone scrolling that feed. Stage kinds are exempt — their uniqueness is in the page, checked on the pixels.
No voiceover field; cues name generated one-shots	See VI.
Hook $\leq$ 7 words; exactly 3 hashtags; ids match weekly position	Pre-existing series rules, unchanged.

VIII · Verification

Nothing reads these videos before they reach an audience. A render that fails halfway still produces a playable MP4, and unattended that mistake repeats four times a day for a week. Five gates, in order, each refusing to let a video past.

Gate	Proves	Runs
1 · Plan validate-plan.mjs	The script is one we would accept: figure named, permitted, complete, numeric where it claims to measure, distinct within its day, no voice.	On every push, before anything renders.
2 · Uniqueness matrix uniqueness-matrix.mjs	No two of the 120 planned videos share a look, a piece of music, or their words. Proved from the plan in about a second.	Before dispatching a render.
3 · File verify-video.mjs	Well-formed 1080×1920 h264, right length, real bitrate, an audio stream with programme in it, non-blank frames, motion that resumes, the end card present.	After the master pass.
4 · Script contract verify-script.mjs	That this file is the video its script asked for. See below.	Immediately after gate 3.
5 · Archive uniqueness.mjs	The finished file's visual and audio fingerprints match nothing already published.	Before upload.

Gate 4, in detail — and the honest limit of it

THE TEMPTING WRONG ANSWER

Look at the pixels and recognise the figure. **It does not work**, and this was measured rather than assumed. The same script rendered before and after the exhibit layer scored **86%** and **81%** on the “how much ink is continuous mark” statistic — the text-only version scored *higher*. Large display type produces runs just as long as a bar does. Erosion and panel-contrast variants separated no better: 4.8% vs 6.3%, and 39% vs 84% with a text-only FounderStory scoring 100%.

A gate that does not separate the two cases is not protecting anything. It is a number that passes, which is worse than no number.

What works is a chain of individually provable steps:

1. The script names a figure that exists, is allowed on that template, and has every field it needs. Proved by reading the plan against the catalogue.
2. The renderer was handed exactly that script. Proved by comparing the props file the render used against the plan item, canonically — key order cannot cause a false failure, and the injected `videoId` is checked against the item id separately because it seeds palette, typeface, motion and musical key.
3. Renders are a pure function of their props. The project already depends on this: a retry producing a different video would invalidate the stored fingerprint.
4. **Therefore the video contains that figure.**

The pixels are still inspected, and still gate — for what they can actually prove. `inspect-band.mjs` decodes only the band and asks whether it is *occupied, alive and held*. That catches what a props comparison cannot: a font that never loaded, a component that threw, a figure that vanished after six frames.

Band check	Floor	Worst real	Catches
Settled ink coverage	3%	6.6%	An empty middle third.
Frames the band changes on	4	26	A figure that is a still image.
Body frames the band is occupied	35%	74%	A figure that flashes rather than holds.

Calibrated against seven real renders across every template. The low end of each is a genuine video, not a bad one — 6.6% is a sparse radar, 74% is a `LogoLadder` clearing its own stage during a scripted freeze. Every floor sits well under the worst real measurement, deliberately: a false positive blocks a publish and needs a human at 3am, while what these catch are gross failures measuring near zero, not near the floor.

STATE IT PLAINLY

The pixel gate proves the band is *occupied and alive*. It does **not** prove which figure is in it. That guarantee comes from the plan contract plus deterministic rendering — a proof, where a classifier would be a guess.

IX · Working practice

Authoring loop

1. Write the script in `plan-source/campaign/w<n>.json`, including `props.exhibit`.
2. `npm run exhibits -- --check` — proposes figures for anything missing one, without writing.
3. `npm run campaign` — rebuilds `plans/` from source.
4. `npm run validate` — all five plan-side gates.
5. `npm run studio` — look at it. Or `npx remotion still src/index.ts <Template> out/probe/x.png --props=<file> --frame=150 --scale=0.4` for a fast single frame.
6. `npm test` — 162 tests, including the catalogue-consistency checks.

Commands

Command	Does
<code>npm run validate</code>	Plan structure, exhibit gate, queue, campaign build freshness, uniqueness matrix.
<code>npm run exhibits [-- --check]</code>	Assign figures to campaign sources, day-aware. <code>--check</code> reports only.
<code>npm run band <file.mp4></code>	Ink, structure, motion and presence in the exhibit band.
<code>npm run verify <file.mp4> [secs]</code>	Container, frames, loudness, end card.
<code>npm run verify:script <mp4> <plan> <id></code>	The full script contract for one finished file.
<code>npm run campaign:check</code>	The 120-video look, music and copy uniqueness matrix.
<code>docs/exhibit-examples.json</code>	Not a command — a validated <code>exhibit</code> block for every kind, ready to paste.

Failure playbook

Message	Do
<code>props.exhibit</code> is missing	Write one, or run <code>npm run exhibits</code> and review what it proposed.
kind “X” is not available to <code><Template></code>	Either the wrong figure for that template, or the catalogue should permit it — decide, then edit <code>registry.json</code> . The consistency test will hold you to it.
must be a finite number on a chart-family figure	The script has no measurement. Use a diagram kind — do not invent a number.
draws the same figure as ... on the same day	Author a different figure for one of them. The assigner already tried most-constrained-first and could not find a clash-free set.
NO FIGURE FITS — write one by hand	Usually labels too long for every candidate. Shorten the lines, or write a figure whose data is not the lines.
the render used different props from the script	The props file is stale. Re-render; do not re-point the check.
the exhibit band is ... % covered	A real render failure — a throw, a font that never loaded, a figure with empty data. Look at a still before anything else.
DIFFERS from its source scripts	Something edited <code>plans/</code> directly. Move the change into <code>plan-source/</code> and rebuild.

X · Appendix — rejected approaches

Recorded so they are not re-attempted, with the measurement that rejected each.

Tried	Measured	Why it was dropped
Run-length “structure” as a text-vs-figure gate	Text-only 86% · exhibit 81%	Text-only scored <i>higher</i> . Large display type produces runs as long as a bar’s.
Erosion (5×5) solidity	4.8% vs 6.3%	Legitimate thin-but-long marks — meters, rules, playheads — erode away with the type.
Panel-contrast detection	39% vs 84%, control 100%	A text-only <code>FounderStory</code> scored 100%. The backdrop vignette dominates the measurement.
Fading the signal colour toward the ground for the ramp	45° hue spread; light end 1.45:1	Not one hue, and its low end was invisible. Replaced by channel scaling with a 0.5 floor.
Re-tinting <code>amber</code> / <code>seaglass</code> to fix CVD	—	Weeks 31–32 are rendered and fingerprinted; a palette edit changes what a re-render of those ids produces.
Greedy left-to-right figure assignment	Two checklists in <code>w36-d05</code>	Cannot back out when a later slot has only one workable figure. Replaced by most-constrained-first.
Keeping the preference order in TypeScript	<code>browser</code> chosen 0 times	The JSON-side tool could not import it, fell back to registry order, and <code>SiteRoast</code> silently lost the stage it was built around.
Radial <code>nodegraph</code> at a 34% orbit	Labels 150px past the panel	A satellite’s label grows outward from its node; the orbit is solved against the panel width, not chosen.
Backfilling <code>plans/</code> directly	5 plans “DIFFER from source”	Generated files. The next campaign build drops every figure written there.
Letting <code>Recap</code> draw <code>bars</code>	Leaderboard on screen twice	The template already draws its leaderboard as real proportional bars. The exhibit’s subject there is the <i>totals</i> , so <code>bars</code> and <code>meters</code> were removed from what <code>Recap</code> may draw.
Requiring a board to have exactly four tiles	<code>Recap</code> fell through to <code>bars</code>	A KPI row of three is a normal form. Three now stack rather than leaving a hole where a fourth should be.

Standing measurements

Quantity	Value
Look space (palette × typeface × motion)	$8 \times 7 \times 6 = 336$
Walk stride (coprime to 336)	149
Campaign videos, all distinct in look and music	120 / 120
Chart pair worst-case CVD separation, all eight palettes	$\Delta E \geq 11.8$
Sequential ramp hue spread, after correction	1°
Ramp light-end contrast, tightest ground (oxblood)	2.30 : 1
Master target / delivered true peak	−23 LUFS / −2 dBTP
Observed mean programme level, mastered	−29.4 dB
Test suite	168 passing

Every rule in this document is enforced by `npm run validate`, `npm test`, or a gate in the render pipeline. If a rule here is not enforced somewhere, it is a bug in the system, not a matter of discipline.